VOICEOPS 2026

ZentroMart Voice Support Agent · Capstone Project Report

Project

The agent answers customer questions from company policies, help pages, and the product catalog. It keeps conversational context, verifies callers before exposing order details, creates human handover tickets, and handles customer audio through Gnani speech services.

Build by step

1 · LLM baseline

A configurable OpenAI-compatible chat endpoint answers practice questions with a support-assistant prompt and no company-document retrieval.

2 · Grounded retrieval

Markdown is split by stable section ID and indexed as TF-IDF vectors alongside catalog rows. Similarity search applies source precedence and downranks superseded policies.

3 · Conversation and tools

The agent retains recent turns and verified order details within a session. The supplied lookup tool checks a phone suffix or delivery pincode and allows one retry. Ticket success is reported only after confirmation.

4 · Voice

Gnani Vachana transcribes customer audio and synthesizes replies with an explicitly named timbre model. Batch output includes transcripts, spoken WAV files, and retrieved source IDs.

Evaluation

Run `python run.py evaluate` and `python run.py evaluate-conversations` with an API key to score prompt-only and retrieval-augmented answers, source-section hits, forbidden claims, required facts, and expected tool calls. Detailed results are saved under `outputs/`.

Measure	Result
Prompt-only fact accuracy	Run evaluator
RAG fact accuracy	Run evaluator
Question retrieval hit rate	Run evaluator
Conversation fact accuracy	Run evaluator
Expected tool-call accuracy	Run evaluator
Forbidden-claim turns	Run evaluator

Lessons and next improvements

The hardest cases are policy precedence, date-sensitive returns, indirect follow-ups, multilingual wording, and privacy around unverified orders. Stable section IDs make evidence auditable, and the lookup tool gates customer details behind verification. Review evaluator misses, tune retrieval, and record the required short demo using your own voice and a Gnani response.